Question	Answer	Marks	AOs		Guidance
10	Curve crosses the x-axis when $y=0$ $y=(k-x)\ln x=0$	**M1**	**3.1a**	Attempt to solve $y=0$	
	Either $k-x=0$ or $\ln x=0$ $x=k$ or 1	**A1**	**1.1b**	Both roots required	
	EITHER Area $=\int_1^k (k-x)\ln x\,\mathrm{d}x$				
	Let $u=\ln x,\ \ \frac{\mathrm{d}v}{\mathrm{d}x}=k-x,\ \ \frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{x},\ \ v=kx-\frac{1}{2}x^2$	**M1**	**2.1**	Using integration by parts with $u=\ln x,\ \ \frac{\mathrm{d}v}{\mathrm{d}x}=k-x$ clearly argued	
	Area $=\left[\left(kx-\frac{1}{2}x^2\right)\ln x\right]_1^k-\int_1^k\frac{1}{x}\left(kx-\frac{1}{2}x^2\right)\mathrm{d}x$	**A1**	**1.1b**	Allow without limits	
	$\left[\left(kx-\frac{1}{2}x^2\right)\ln x\right]_1^k-\int_1^k\left(k-\frac{1}{2}x\right)\mathrm{d}x$	**M1**	**3.1a**	Simplifying the integrand	
	$\left[\left(kx-\frac{1}{2}x^2\right)\ln x-\left(kx-\frac{1}{4}x^2\right)\right]_1^k$	**A1**	**1.1b**	Second part correct	
	$\left(\left(k^2-\frac{1}{2}k^2\right)\ln k-\left(k^2-\frac{1}{4}k^2\right)\right)-\left(\left(k-\frac{1}{2}\right)\ln 1-\left(k-\frac{1}{4}\right)\right)$	**M1dep**	**1.1a**	Using limits. Dependendent on M mark for integration by parts	
	$=\frac{1}{2}k^2\ln k-\frac{3}{4}k^2+k-\frac{1}{4}$	**A1** **[8]**	**1.1b**	Cao	

Question	Answer	Marks	AOs		Guidance
10	OR Integral split into two separate integrals $\int_1^k k\ln x\ \mathrm{d}x$				
	Let $u=\ln x,\ \ \frac{\mathrm{d}v}{\mathrm{d}x}=k,\ \ \frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{x},\ \ v=kx$	**M1**		Using integration by parts with $u=\ln x,\ \ \frac{\mathrm{d}v}{\mathrm{d}x}=k$ or $u=\ln x,\ \ \frac{\mathrm{d}v}{\mathrm{d}x}=\pm x$ clearly argued	
	$=\left[kx\ln x\right]_1^k-\int_1^k\frac{1}{x}kx\ \mathrm{d}x$	**M1**		Simplifying the integrand	
	$\left[kx\ln x\right]_1^k-\int_1^k k\ \mathrm{d}x=\left[kx\ln x-kx\right]_1^k$ $\left(k^2\ln k-k^2\right)-\left(k\ln 1-k\right)=k^2\ln k-k^2+k$	**M1dep**		Substituition of limits seen in at least one integral. Dependendent on M mark for integration by parts	
	And Area $=\int_1^k x\ln x\ \mathrm{d}x$				
	Let $u=\ln x,\ \ \frac{\mathrm{d}v}{\mathrm{d}x}=x,\ \ \frac{\mathrm{d}u}{\mathrm{d}x}=\frac{1}{x},\ \ v=\frac{1}{2}x^2$				
	$=\left[\frac{1}{2}x^2\ln x\right]_1^k-\int_1^k\frac{1}{x}\times\frac{1}{2}x^2\mathrm{d}x$	**A1**		Both integrals correct at this stage Allow without limits	
	$\left[\frac{1}{2}x^2\ln x\right]_1^k-\int_1^k\frac{1}{2}x\ \mathrm{d}x$ $\left[\frac{1}{2}x^2\ln x-\frac{1}{4}x^2\right]_1^k$	**A1**		Both integrals fully correct Allow without limits	
	$\left(\frac{1}{2}k^2\ln k-\frac{1}{4}k^2\right)-\left(\frac{1}{2}\ln 1-\frac{1}{4}\right)=\frac{1}{2}k^2\ln k-\frac{1}{4}k^2+\frac{1}{4}$ Area $=\left(k^2\ln k-k^2+k\right)-\left(\frac{1}{2}k^2\ln k-\frac{1}{4}k^2+\frac{1}{4}\right)$				
	$=\frac{1}{2}k^2\ln k-\frac{3}{4}k^2+k-\frac{1}{4}$	**A1**		Cao	